

SHETH L.U.J. AND SIR M.V. DEGREE COLLEGE OF SCIENCE

SUBJECT NAME: S101 Data Analysis with SAS / SPSS / R Practical

PRACTICAL - 12

AIM : Write code to Combining datasets vertically (concatenation) using rbind() in R studio.

Input :

```
# PRACTICAL 12 – Combining datasets vertically using rbind()
library(dplyr)
# 1. IMPORT BOTH DATASETS
usage1 <- read.csv("random_smartphone_usage_dataset.csv",
  na.strings = c("", "NA"),
  stringsAsFactors = FALSE)
usage2 <- read.csv("mobile_usage_behavioral_analysis.csv",
  na.strings = c("", "NA"),
  stringsAsFactors = FALSE)
cat("\n--- Dataset 1 Columns ---\n")
print(names(usage1))
cat("\n--- Dataset 2 Columns ---\n")
print(names(usage2))

# 2. FIND COMMON COLUMNS SAFELY
common_cols <- intersect(names(usage1), names(usage2))
cat("\n--- Common Columns Detected ---\n")
print(common_cols)
# If NO common columns are found, stop safely
if (length(common_cols) == 0) {
  stop(" ERROR: No common columns found between the two datasets. rbind() cannot
be applied.")
}

# 3. ALIGN BOTH DATASETS USING ONLY COMMON COLUMNS
usage1_clean <- usage1[, common_cols, drop = FALSE]
usage2_clean <- usage2[, common_cols, drop = FALSE]

# 4. ENSURE DATA TYPES MATCH (SAFE LOOP)
for (col in common_cols) {
```

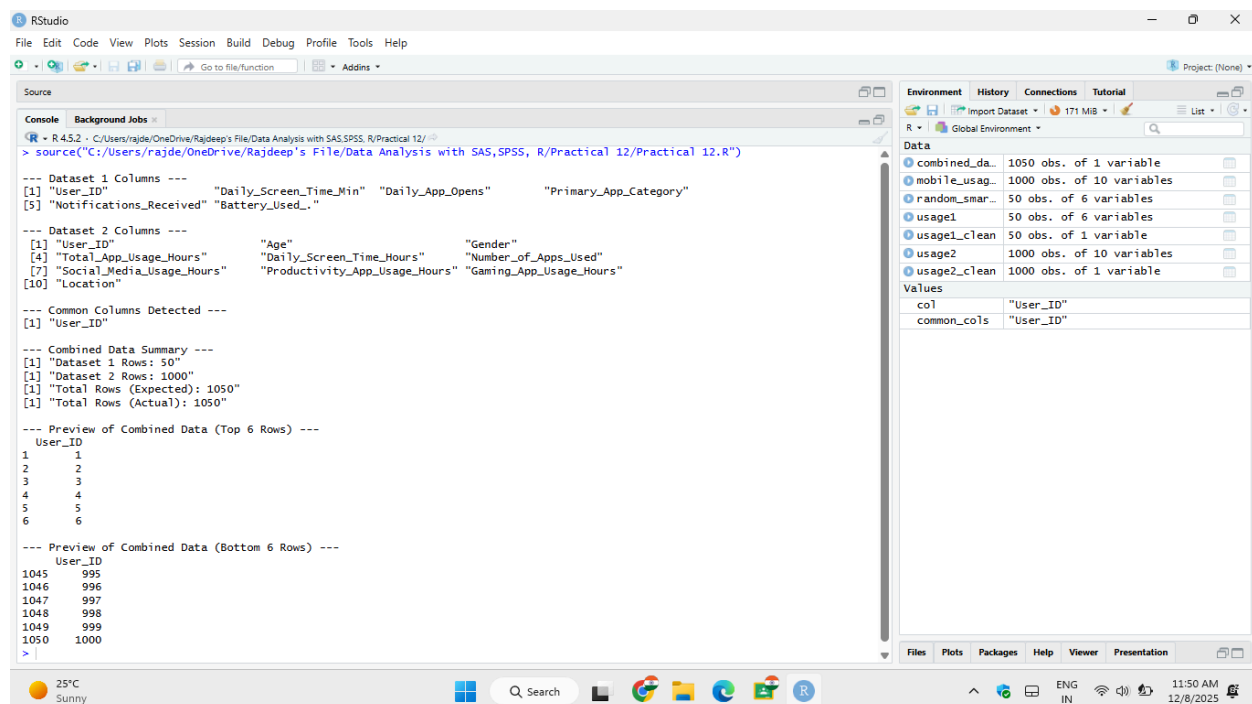
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```
# If either column is numeric, force both to numeric
if (is.numeric(usage1_clean[[col]]) || is.numeric(usage2_clean[[col]])) {
  usage1_clean[[col]] <- as.numeric(usage1_clean[[col]])
  usage2_clean[[col]] <- as.numeric(usage2_clean[[col]])
}
}

# 5. VERTICAL COMBINATION USING rbind()
combined_data <- rbind(usage1_clean, usage2_clean)
cat("\n--- Combined Data Summary ---\n")
print(paste("Dataset 1 Rows:", nrow(usage1_clean)))
print(paste("Dataset 2 Rows:", nrow(usage2_clean)))
print(paste("Total Rows (Expected):", nrow(usage1_clean) + nrow(usage2_clean)))
print(paste("Total Rows (Actual):", nrow(combined_data)))
cat("\n--- Preview of Combined Data (Top 6 Rows) ---\n")
print(head(combined_data))
cat("\n--- Preview of Combined Data (Bottom 6 Rows) ---\n")
print(tail(combined_data))
```

Output :



The screenshot shows the RStudio interface. The console window displays the following output:

```
> source("C:/Users/rajde/OneDrive/Rajdeep's File/Data Analysis with SAS,SPSS, R/Practical 12/Practical 12.R")

--- Dataset 1 Columns ---
[1] "User_ID" "Daily_Screen_Time_Min" "Daily_App_Opens" "Primary_App_Category"
[5] "Notifications_Received" "Battery_Used_..."

--- Dataset 2 Columns ---
[1] "User_ID" "Age" "Gender"
[4] "Total_App_Usage_Hours" "Daily_Screen_Time_Hours" "Number_of_Apps_Used"
[7] "Social_Media_Usage_Hours" "Productivity_App_Usage_Hours" "Gaming_App_Usage_Hours"
[10] "Location"

--- Common Columns Detected ---
[1] "User_ID"

--- Combined Data Summary ---
[1] "Dataset 1 Rows: 50"
[1] "Dataset 2 Rows: 1000"
[1] "Total Rows (Expected): 1050"
[1] "Total Rows (Actual): 1050"

--- Preview of Combined Data (Top 6 Rows) ---
  User_ID
1       1
2       2
3       3
4       4
5       5
6       6

--- Preview of Combined Data (Bottom 6 Rows) ---
  User_ID
1045    995
1046    996
1047    997
1048    998
1049    999
1050   1000
```

The Environment window on the right shows the following data objects:

Object	Size
combined_da...	1050 obs. of 1 variable
mobile_usag...	1000 obs. of 10 variables
random_smar...	50 obs. of 6 variables
usage1	50 obs. of 6 variables
usage1_clean	50 obs. of 1 variable
usage2	1000 obs. of 10 variables
usage2_clean	1000 obs. of 1 variable

The Values window shows the common columns:

col	Value
common_cols	"User_ID"